METRIC
MIL-STD-3072
23 January 2023

DEPARTMENT OF DEFENSE
INTERFACE STANDARD

GROUND VEHICLE 600 VOLTS DC ELECTRIC POWER CHARACTERISTICS



AMSC N/A

FSC 2920

DISTRIBUTION A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by all departments and agencies of the Department of Defense (DoD).

2. The intent of this document is to ensure compatibility between ground vehicle 600 VDC electric systems, 600 VDC external power, and onboard 600 VDC utilization equipment.

3. Comments, suggestions, or questions on this document should be addressed to U.S. Army Ground Vehicle Systems Center, ATTN: FCDD-GVS-SAT MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000, or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
FOREWORD		ii
1.	SCOPE	1
1.1	Scope	1
2.	APPLICABLE DOCUMENTS	1
2.1	General	1
2.2	Government Documents	1
2.2.1	Specifications, standards, and handbooks	1
2.3	Order of precedence	1
3.	DEFINITIONS	2
3.1	Acronyms	2
3.2	Distortion factor	2
3.3	Distortion spectrum	2
3.4	Operations	2
3.4.1	Abnormal operation	2
3.4.2	Normal operation	2
3.5	Steady state voltage	3
3.6	Utilization equipment	3
4.	GENERAL REQUIREMENTS	3
4.1	Ground vehicle 600 VDC power system	3
4.1.1	Ground vehicle 600 VDC power source performance	3
4.1.2	Ground vehicle 600 VDC power system performance	3
4.2	Ground vehicle 600 VDC utilization equipment requirements	3
4.2.1	Power compatibility	3
4.2.2	Operation	3
4.2.3	Power failure	4
4.2.4	External power source requirement	4
4.2.5	Test requirement	4
5.	DETAILED REQUIREMENTS	4
5.1	Power characteristics	4
5.1.1	System type	4
5.1.2	Normal operation	4
5.1.3	Abnormal operation	4
5.2	Load characteristics	4
5.2.1	Grounding	5
5.2.2	Polarity	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

PARAGRAPH PAGE

6. NOTES 5
6.1 Intended use 5
6.2 Acquisition requirements 5
6.3 Subject term (key word) listing..... 5

TABLE PAGE

I. 600 VDC normal operation characteristics. 5

FIGURE PAGE

1. Envelope of normal transients and abnormal operation for 600 VDC system. 6
2. Maximum distortion spectrum for 600 VDC system..... 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard establishes the requirements and characteristics of ground vehicle 600 VDC electric power provided at the input terminals of electric utilization equipment. A forthcoming handbook will define test methods and procedures for determining onboard utilization equipment compatibility with the electric power characteristics requirements defined herein. A forthcoming system performance specification will define the interface, behavior, and safety features of a 600 VDC power system for military ground vehicles. Electromagnetic interference and voltage spikes are not covered by this standard.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, the document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 and 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

(Copies of these documents are available from https://quicksearch.dla.mil/.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this standard takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Acronyms.

AMSC	-	Acquisition Method Suffix Code
DoD	-	Department of Defense
DC	-	Direct Current
FSC	-	Federal Supply Code
ms	-	milliseconds
NHSS	-	Normal High Steady State
NLSS	-	Normal Low Steady State
RMS	-	Root Mean Square
s	-	seconds
STANAG	-	Standardization Agreement
UUT	-	Unit Under Test
VDC	-	Volts Direct Current

3.2 Distortion factor. The ratio of the root mean square (RMS) DC voltage distortion to the DC steady state voltage.

3.3 Distortion spectrum. Quantifies RMS voltage distortion in terms of the amplitude of each frequency component. The distortion spectrum includes the components resulting from amplitude and frequency modulation as well as harmonic and nonharmonic components of the waveform.

3.4 Operations. When supplied electric power having characteristics specified herein, utilization equipment provides the level of performance required by the system or component specification for each operating condition. Operation of utilization equipment will not cause the power characteristics at its terminals to exceed the limits specified herein when tested as specified in the forthcoming handbook to accompany this standard. The electrical system will not be adversely affected by the operation of compliant utilization equipment.

3.4.1 Abnormal operation. When a malfunction or failure in the electric system has taken place and the protective devices of the system are operating to remove the malfunction or failure from the remainder of the system before the limits for abnormal operation are exceeded. The power source may operate in a degraded mode on a continuous basis where the power characteristics supplied to the utilization equipment are outside normal operation limits but remain within the limits for abnormal operation.

3.4.2 Normal operation. When the system is operating as intended in the absence of any fault or malfunction that degrades performance beyond established requirements. It includes all system functions required for vehicle operation. Normal operation includes switching of utilization equipment, main engine speed changes, main engine starting operation, synchronizing, and paralleling of power sources, and operation from external power sources.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

3.5 Steady state voltage. Mean voltage, including ripple, measured over a period of more than 100ms and less than one second.

3.6 Utilization equipment. The electronic device, equipment, or system subjected to the voltage range(s) indicated in this specification.

4. GENERAL REQUIREMENTS

4.1 Ground vehicle 600 VDC power system.

4.1.1 Ground vehicle 600 VDC power source performance. The characteristics of the electrical power measured at the output terminals of an unregulated power source or the point of regulation of a regulated power source shall be within the limits specified in the applicable specification of the power source.

4.1.2 Ground vehicle 600 VDC power system performance. The ground vehicle 600 VDC power system shall provide electric power having the characteristics specified in this standard at the utilization equipment terminals during all operations of the power system. It is the responsibility of the vehicle's design authority to provide the distribution and protection network to the terminals of the utilization equipment while maintaining the power characteristics specified in this standard.

4.2 Ground vehicle 600 VDC utilization equipment requirements.

4.2.1 Power compatibility. Utilization equipment shall be compatible with the power characteristics specified herein. Utilization equipment shall require electric power of no better quality than specified herein. Utilization equipment shall be compatible with the applicable ground vehicle specification requirements for control of electromagnetic interference and voltage spikes induced by lightning, electromagnetic pulses, and power switching. Electromagnetic interference and voltage spikes are not covered by this standard, but applicable parts of MIL-STD-461 shall be used to determine that the utilization equipment complies with this standard.

4.2.2 Operation. When supplied electric power having characteristics specified herein, vehicle utilization equipment shall provide the level of performance required by the system or component specification for each operating condition. Operation of utilization equipment shall not cause the power characteristics at its terminals to exceed the limits specified herein. Utilization equipment operation under any electric system operating condition shall not have an adverse effect on or cause malfunction of the electric system.

4.2.2.1 Normal operation. Utilization equipment shall provide the level of performance specified in the system or component specification.

4.2.2.2 Abnormal operation. Utilization equipment shall be permitted a degradation or loss of function unless otherwise specified in the system or component specification. Utilization

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

equipment shall not suffer damage or cause an unsafe condition. Utilization equipment shall resume full performance when normal operation of the electrical system is restored.

4.2.3 Power failure. The loss of power to any utilization equipment terminal shall not result in an unsafe condition or damage to utilization equipment.

4.2.4 External power source requirement. External electric power sources shall supply power having the characteristics specified in this standard at the power input terminals of the onboard utilization equipment. In order to allow for steady state voltage drop between the vehicle external power receptacle and the onboard utilization equipment terminals, the voltage at the external power receptacle shall be 575 to 635 volts (DC voltage drop of 0 to 10 volts).

4.2.5 Test requirement. Equipment testing is required to demonstrate utilization equipment compatibility with the electric power characteristics of this standard. The applicable test methods of the forthcoming handbook to accompany this standard and MIL-STD-461 shall be used to determine that the utilization equipment complies with this standard. Utilization equipment test requirements shall be in accordance with the equipment's specification. Ground vehicles shall be tested to demonstrate that the vehicle electric system power characteristics are within the limits of this standard throughout all operating conditions of the vehicle and its utilization equipment. Vehicle test requirements shall be in accordance with the system specification.

4.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. DETAILED REQUIREMENTS

5.1 Power characteristics.

5.1.1 System type. The power system shall provide power using direct current via paired wires, source and return, having a nominal voltage of 600 Volts. Power at the input of the utilization equipment shall conform to figure 1, figure 2, and table I.

5.1.2 Normal operation. Normal operation characteristics shall be in accordance with figure 1, figure 2, and table I.

5.1.3 Abnormal operation. Overvoltage and undervoltage values shall be within the limits of figure 1.

5.2 Load characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

5.2.1 Grounding. All electrical power input terminals, including negative terminals, shall be galvanically isolated from case ground by a minimum of 100MΩ during normal operation. Equipment chassis shall not be used for power returns.

5.2.2 Polarity. Equipment shall not be damaged by reversal of positive and negative connections. Employing a positive physical means to prevent polarity reversal shall also fulfill this requirement.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intent of this document is to define the 600 VDC electrical power quality requirements for military ground vehicle systems and utilization equipment. This lays the groundwork for commonality across vehicle platforms. The manufacturer of utilization equipment is responsible for its compliance with these requirements. The vehicle's design authority is responsible to ensure that the 600 VDC delivered to the input power terminal of the utilization equipment meets these requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

6.3 Subject term (key word) listing.

DC voltage
Voltage
High Voltage

TABLE I. 600 VDC normal operation characteristics.

Steady-state Voltage	565 – 635 V (600±35 V)
Normal Transients	475 – 725 V for 15 ms (see figure 1)
Ripple Amplitude	9.0 V maximum
Distortion Factor	0.011 maximum
Distortion Spectrum	See figure 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

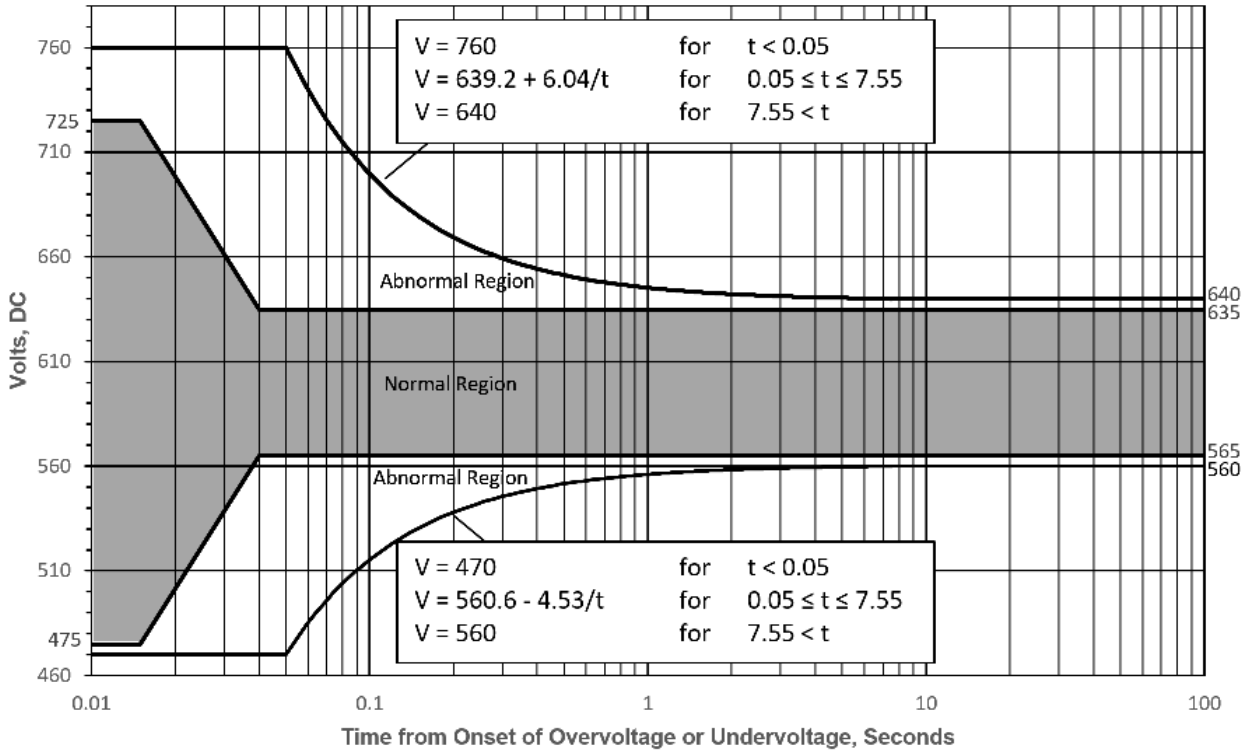


FIGURE 1. Envelope of normal transients and abnormal operation for 600 VDC system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

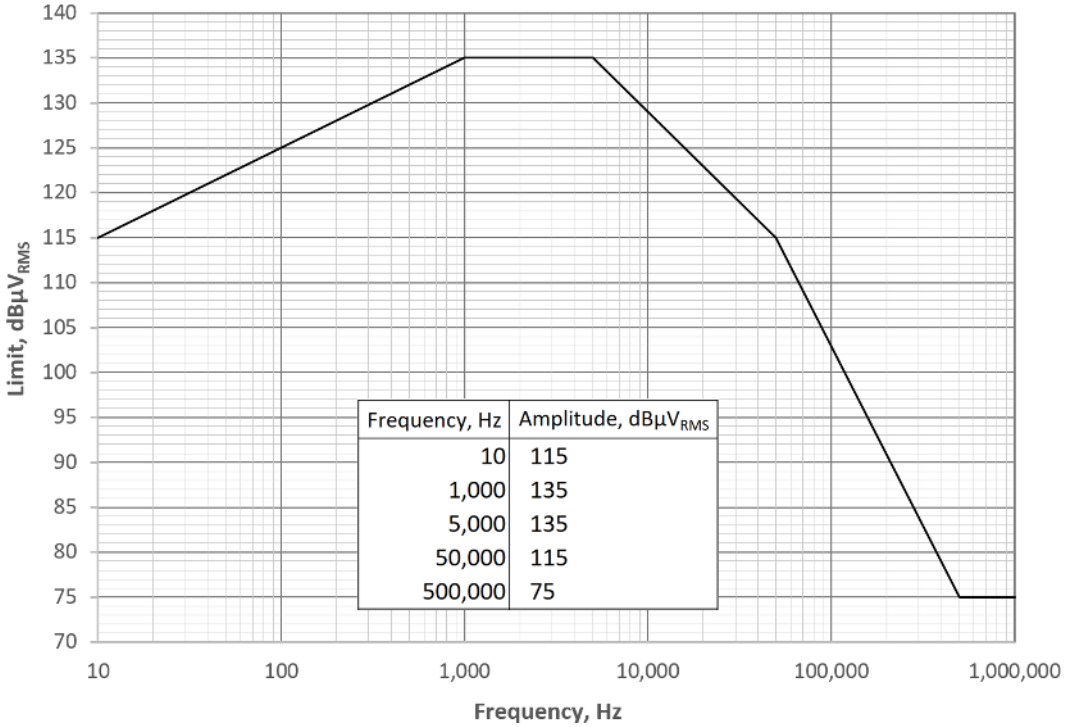


FIGURE 2. Maximum distortion spectrum for 600 VDC system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
DLA – CC

Preparing Activity:
Army - AT
(Project 2920-2022-001)

Review Activities:
Army – AV, CR, MI, TE
Navy – YD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online, database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.